Nonhomogeneous Equations & Undetermined Coefficients

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Consider the differential equation $y'' - 3y' + 2y = f(t)$ with initial conditions $y(0) = 0$ and $y'(0) = 0$. If the forcing function is $f(t) = u(t-2)$, where $u(t)$ is the Heaviside step function, what is the Laplace transform of the solution $Y(s) = \mathcal{L}\{y(t)\}$?

The Laplace transform of the differential equation is $s^2Y(s) - sy(0) - y'(0) - 3(sY(s) - y(0)) + 2Y(s) = \mathcal{L}\{f(t)\}$. Given $y(0) = 0$ and $y'(0) = 0$, this simplifies to $s^2Y(s) - 3sY(s) + 2Y(s) = \mathcal{L}\{u(t-2)\}$, so $Y(s)(s^2 - 3s + 2) = \mathcal{L}\{u(t-2)\}$. The Laplace transform of $u(t-a)$ is $\frac{e^{-as}}{s}$. Therefore, $\mathcal{L}\{u(t-2)\} = \frac{e^{-2s}}{s}$. Thus, $Y(s) = \frac{e^{-2s}}{s(s^2 - 3s + 2)}$. Factoring the denominator, $s^2 - 3s + 2 = (s-1)(s-2)$. So.

2 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Construct the Green's kernel $G(x,t)$ for $L[y] = y'' - y$, and express the particular solution of $y'' - y = \frac{1}{e^x + 1}$ satisfying $y(0) = 0, y'(0) = 0$ as an explicit definite integral.

Step 1: Basis & Wronskian: $y_1(x) = e^x$, $y_2(x) = e^{-x}$. $W(t) = e^t(-e^{-t}) - e^t(e^{-t}) = -2$. Step 2: Green's kernel: $G(x,t) = \frac{y_1(t)y_2(x) - y_1(x)y_2(t)}{W(t)} = \frac{e^te^{-x} - e^xe^{-t}}{-2} = \frac{e^{x-t} - e^{-(x-t)}}{2} = \sinh(x-t)$. Step 3: Integral representation: $y(x) = \int_0^x \sinh(x-t) \cdot \frac{1}{e^t + 1}\,dt$. Verifying: $G(t,t) = \sinh(0) = 0$, and $\frac{\partial G}{\partial x}\big|_{x=t} = \cosh(0) = 1$, satisfying the universal Cauchy impulse propert.

3 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Find a particular solution of $y'' + 4y = \sec(2x)$ for $x \in (-\pi/4, \pi/4)$ using Variation of Parameters.

Step 1: Basis & Wronskian: $y_1 = \cos 2x$, $y_2 = \sin 2x$. $W = \cos 2x(2\cos 2x) - (-2\sin 2x)\sin 2x = 2$. Step 2: Parameter integrals: $u_1'(x) = -\frac{y_2(x)g(x)}{W} = -\frac{\sin 2x \sec 2x}{2} = -\frac{1}{2}\tan 2x \Rightarrow u_1(x) = \frac{1}{4}\ln|\cos 2x|$. $u_2'(x) = \frac{y_1(x)g(x)}{W} = \frac{\cos 2x \sec 2x}{2} = \frac{1}{2} \Rightarrow u_2(x) = \frac{x}{2}$. Step 3: Particular solution: $y_p(x) = u_1y_1 + u_2y_2 = \frac{1}{4}\cos(2x)\ln|\cos 2x| + \frac{x}{2}\sin(2x)$.

4 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Solve the initial value problem with step forcing using the Laplace transform:

$y'' + 4y = 8u(t-2), \qquad y(0) = 0, \quad y'(0) = 0.$

Step 1: Transform equation: $(s^2 + 4)Y(s) = \frac{8e^{-2s}}{s} \Rightarrow Y(s) = 8e^{-2s} \cdot \frac{1}{s(s^2 + 4)}$. Step 2: Partial fractions on transfer factor: $\frac{1}{s(s^2 + 4)} = \frac{1/4}{s} - \frac{s/4}{s^2 + 4} \Rightarrow \mathcal{L}^{-1}\left\{\frac{8}{s(s^2 + 4)}\right\} = 2 - 2\cos(2t)$. Step 3: Apply Second Shift Theorem: $y(t) = u(t-2)(2 - 2\cos(2(t-2)))$.

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5 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the differential equation $\frac{dy}{dx} = x + y$ with the initial condition $y(0) = 1$. You are asked to approximate $y(0.2)$ using Euler's method. If the step size $h = 0.1$ is used, and it is known that the exact solution at $x = 0.2$ is $y(0.2) = 1.24$, determine the value of k such that $y(0.2)_{\text{Euler}} = 1.24 - k$.

Euler's method for approximating the solution to $\frac{dy}{dx} = f(x, y)$ with initial condition $y(x_0) = y_0$ and step size h is given by $y_{n+1} = y_n + hf(x_n, y_n)$. In this problem, $f(x, y) = x + y$, $x_0 = 0$, $y_0 = 1$, and $h = 0.1$. We want to approximate $y(0.2)$ with $h = 0.1$. This requires two steps: Step 1: From $x_0 = 0$ to $x_1 = 0.1$. $y_1 = y_0 + hf(x_0, y_0)$ $y_1 = 1 + 0.1(0 + 1)$ $y_1 = 1 + 0.1(1)$ $y_1 = 1 + 0.1$ $y_1 = 1.1$ Step 2: From $x_1 = 0.1$ to $x_2 = 0.2$..

6 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the coefficient a_5 of the power series solution $y = \sum a_n x^n$ of $y'' - xy' - y = 0$ with $y(0) = 1$, $y'(0) = 2$.

Substitute the series and match powers of x to get a recurrence; with $a_0 = 1,\ a_1 = 2$ the coefficients are $a_0 = 1, a_1 = 2, a_2 = \frac{1}{2}, a_3 = \frac{2}{3}, a_4 = \frac{1}{8}, a_5 = \frac{2}{15}$. Final Answer: $\frac{2}{15}$

7 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A harvested population evolves according to $\frac{dP}{dt} = P\left(1 - \frac{P}{10}\right) - 2.1$. Determine the two equilibrium populations, identify which is the extinction threshold, and determine whether a starting population $P(0) = 2$ survives or collapses.

Step 1: Equilibrium quadratic: Set $\frac{dP}{dt} = 0$: $-\frac{P^2}{10} + P - 2.1 = 0 \Rightarrow P^2 - 10P + 21 = 0.$ Factoring: $(P - 3)(P - 7) = 0 \Rightarrow P_1 = 3,\ P_2 = 7$. Step 2: Stability: $f'(P) = 1 - \frac{P}{5}$. $f'(3) = 1 - 0.6 = +0.4 > 0 \Rightarrow P = 3$ is an unstable extinction threshold. $f'(7) = 1 - 1.4 = -0.4 < 0 \Rightarrow P = 7$ is the stable carrying capacity. Step 3: Dynamics for $P(0) = 2$: Since $P(0) = 2 < 3$, $\frac{dP}{dt} < 0$. The population strictly declines and collapses to.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Solve $y'' + 4y' + 4y = 0$ with $y(0) = 2$ and $y'(0) = 1$. Determine the exact time $t > 0$ at which $y(t)$ attains its absolute maximum value.

Step 1: Characteristic roots: $r^2 + 4r + 4 = (r+2)^2 = 0 \Rightarrow r = -2$ (repeated). General solution: $y(t) = (C_1 + C_2 t)e^{-2t}$. Step 2: Initial conditions: $y(0) = C_1 = 2$.

$y'(t) = C_2 e^{-2t} - 2(C_1 + C_2 t)e^{-2t} = ((C_2 - 2C_1) - 2C_2 t)e^{-2t}$. $y'(0) = C_2 - 2(2) = 1 \Rightarrow C_2 = 5$. Particular solution: $y(t) = (2 + 5t)e^{-2t}$. Step 3: Find maximum: Set $y'(t) = 0$:

$((5-4) - 10t)e^{-2t} = (1 - 10t)e^{-2t} = 0 \Rightarrow 1 - 10t = 0 \Rightarrow t_{\max} = \frac{1}{10} = 0.1$. Maximu.

9 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Consider the stiff decay equation $\frac{dy}{dt} = -100y$ with $y(0) = 1$. Determine the stability threshold h_{crit} for Forward Euler, and compute the value of $y(0.1)$ using Backward (Implicit) Euler with step size $h = 0.05$.

Step 1: Forward Euler stability threshold: For $y' = \lambda y$ with $\lambda = -100$, the amplification factor is $1 + h\lambda = 1 - 100h$. Stability requires: $|1 - 100h| < 1 \Rightarrow -1 < 1 - 100h < 1 \Rightarrow h < \frac{2}{100} = 0.02$. Thus $h_{\text{crit}} = 0.02$. If $h > 0.02$, Forward Euler explodes uncontrollably. Step 2: Backward (Implicit) Euler formula:

$y_{n+1} = y_n + h(-100y_{n+1}) \Rightarrow (1 + 100h)y_{n+1} = y_n \Rightarrow y_{n+1} = \frac{1}{1+100h} y_n$. With $h = 0.05$.

10 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Use the Laplace transform and the Convolution Theorem to solve the integral equation $y(t) = t^2 + \int_0^t e^{t-\tau} y(\tau)\, d\tau$.

Step 1: Recognize convolution: The equation is $y(t) = t^2 + e^t * y(t)$. Step 2: Transform both sides: $Y(s) = \frac{2}{s^3} + \left(\frac{1}{s-1}\right)Y(s) \Rightarrow Y(s)\left(1 - \frac{1}{s-1}\right) = \frac{2}{s^3}$. $Y(s)\left(\frac{s-2}{s-1}\right) = \frac{2}{s^3} \Rightarrow Y(s) = \frac{2(s-1)}{s^3(s-2)}$. Step 3: Partial fractions: $\frac{2(s-1)}{s^3(s-2)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s^3} + \frac{D}{s-2}$. By cover-up, $D = \frac{2(2-1)}{2^3} = \frac{2}{8} = \frac{1}{4}$.

11 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Consider the initial value problem $\frac{dy}{dx} = x + y$ with $y(0) = 1$. Use the Euler method with step size $h = 0.5$ to approximate $y(1)$.

The Euler method approximates the solution of an ODE $\frac{dy}{dx} = f(x, y)$ with initial condition $y(x_0) = y_0$ using the iterative formula $y_{i+1} = y_i + hf(x_i, y_i)$, where h is the step size and $x_i = x_0 + ih$. Given: $f(x, y) = x + y$, $x_0 = 0$, $y_0 = 1$, and $h = 0.5$. We want to approximate $y(1)$. Since $1 = 0 + 2 \times 0.5$, we need two steps. Step 1: Approximate $y(0.5)$. $x_1 = x_0 + h = 0 + 0.5 = 0.5$

$y_1 = y_0 + hf(x_0, y_0) = 1 + 0.5(0 + 1) = 1 + 0.5(1) = 1.5$. Step 2: Approximate.

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12 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Solve $\frac{dy}{dx} = \frac{1+y^2}{x^2-4}$ on $(-2, 2)$ with $y(0) = 1$. At what value of x does $y(x) \to +\infty$?

Separable: $\frac{dy}{1+y^2} = \frac{dx}{x^2-4} \Rightarrow \arctan y = \frac{1}{4}\ln\left(\frac{2-x}{2+x}\right) + C$. Using $y(0) = 1$ gives $\arctan(1) = \frac{\pi}{4} = 0 + C \Rightarrow C = \frac{\pi}{4}$. Blow-up $y \to \infty$ occurs when $\arctan y \to \frac{\pi}{2}$, so $\frac{1}{4}\ln\left(\frac{2-x}{2+x}\right) + \frac{\pi}{4} = \frac{\pi}{2} \Rightarrow \ln\left(\frac{2-x}{2+x}\right) = \pi \Rightarrow \frac{2-x}{2+x} = e^{\pi}$. Solving for x.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Solve $x^2y' + xy = \frac{1}{x^2+1}$ for $x > 0$, with $y(1) = \frac{\pi}{4}$. What is $y(\sqrt{3})$?

Standard form: $y' + \frac{1}{x}y = \frac{1}{x(x^2+1)}$. $\mu = x$. Multiplying gives $(xy)' = \frac{1}{x(x^2+1)} = \frac{1}{x} - \frac{x}{x^2+1}$. Integrating gives $xy = \ln x - \frac{1}{2}\ln(x^2+1) + C$. Using $y(1) = \pi/4$ yields $\pi/4 = -\frac{1}{2}\ln 2 + C \Rightarrow C = \frac{\pi}{4} + \frac{1}{2}\ln 2$. At $x = \sqrt{3}$, $\sqrt{3}y = \frac{1}{2}\ln 3 - \frac{1}{2}\ln 4 + \frac{\pi}{4} + \frac{1}{2}\ln 2 = \frac{\pi}{4} + \frac{1}{2}\ln(3/2)$. Thus.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

[Nonhomogeneous Equations, Undetermined Coefficients & Resonance · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Linear First-Order Equations & the Integrating Factor

Level 1 · FoundationsOn an AP Calculus BC exam question, a student writes $\int_0^5 (R(t) - D(t))\,dt = 42.18$ as their final response to a question asking for the total amount of water in a tank at $t = 5$, given initial volume 100 gallons. Identify the missing component in the student's solution.

The student computed only the net change $\int_0^5 (R(t) - D(t))\,dt$, failing to add the initial condition $W(0) = 100$. By the Fundamental Theorem of Calculus: $W(5) = W(0) + \int_0^5 (R(t) - D(t))\,dt = 100 + 42.18 = \mathbf{142.18}$ gallons. Failing to include the initial condition is a common fatal error on FRQ Question 1.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 3.01 Second-Order Linear Homogeneous Equations

Level 3 · Advanced / AP FRQLet $f(x) = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}$. Identify the closed-form elementary function that this series represents, and determine its radius of convergence.

Write out the terms: $f(x) = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \frac{x^7}{7!} + \ldots$ Recall the definition of the hyperbolic sine function: $\sinh(x) = \frac{e^x - e^{-x}}{2}$. Expanding e^x and e^{-x} shows that only odd powers remain with coefficients $1/(2n+1)!$. Thus $f(x) = \sinh\mathbf{(x)}$. Apply the Ratio Test: \[\lim_{n\to\infty} \frac{x^{2n+3}}{(2n+3)!} \cdot \frac{(2n+1)!}{x^{2n+1}} = x^2 \lim_{n\to\infty} \frac{1}{(2n+3)(2n+2)} = 0

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17 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)** • **Cross-Lesson: 5.01 Variation of Parameters**

Level 2 · MediumOn Section II Part A (Calculator Active), a student is asked for the acceleration vector at $t = 3$ for a particle whose velocity is $\vec{v}(t) = \langle \ln(t^2 + 1), 3\cos(t) \rangle$. What calculator command must be used to find this vector, and what is its 3-decimal numerical value?

Acceleration is the derivative of velocity: $\vec{a}(3) = \langle x''(3), y''(3) \rangle$. Use the calculator's numerical derivative function nDeriv: • $x''(3) = \frac{d}{dt}[\ln(t^2 + 1)]\Big|_{t=3} = \frac{2(3)}{3^2+1} = \frac{6}{10} = 0.6$. • $y''(3) = \frac{d}{dt}[3\cos(t)]\Big|_{t=3} = -3\sin(3) \approx -0.423$. Thus: $\mathbf{\vec{a}(3) = \langle 0.600, -0.423 \rangle}$.

18 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)** • **Cross-Lesson: 1.01 Separable Equations**

[AP Calculus BC Free-Response Questions (FRQs) · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

19 **[Challenging]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)** • **Cross-Lesson: 2.01 Linear First-Order Equations & the Integrating Factor**

Apply two steps of Heun's method (Improved Euler) with step size $h = 0.1$ to approximate $y(0.2)$ for the IVP $\frac{dy}{dt} = 2ty, \qquad y(0) = 1.$ Compare your approximation with the exact analytical value $y(0.2) = e^{(0.2)^2} = e^{0.04} \approx 1.04081077$.

Step 1: First interval $[0, 0.1]$: Initial point: $t_0 = 0, y_0 = 1$. Slope at start: $k_1 = f(0, 1) = 2(0)(1) = 0$. Predictor: $\tilde{y}_1 = y_0 + hk_1 = 1 + 0.1(0) = 1$. Slope at endpoint: $k_2 = f(0.1, 1) = 2(0.1)(1) = 0.2$. Corrector: $y_1 = y_0 + \frac{h}{2}(k_1 + k_2) = 1 + \frac{0.1}{2}(0 + 0.2) = 1 + 0.01 = 1.0100.$ Step 2: Second interval $[0.1, 0.2]$: Point: $t_1 = 0.1, y_1 = 1.01$. Slope at start: $k_1 = f(0.1, 1.01) = 2(0.1)(1.01) = 0.2020$. Predictor.

20 **[Challenging]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)** • **Cross-Lesson: 3.01 Second-Order Linear Homogeneous Equations**

An RC circuit with resistance $R = 2\,\Omega$ and capacitance $C = 0.5$ F is driven by a delayed step voltage $E(t) = 10u(t - 2)$. With $q(0) = 0$, find the exact charge $q(t)$ for $t \geq 0$.

Step 1: Governing ODE: $R\frac{dq}{dt} + \frac{1}{C}q = E(t) \Rightarrow 2q' + 2q = 10u(t-2) \Rightarrow q' + q = 5u(t-2)$. Step 2: Integrating factor: $\mu(t) = e^t$. For $0 \leq t < 2$, $E(t) = 0 \Rightarrow q(t) = 0$. For $t \geq 2$: $\frac{d}{dt}[e^t q] = 5e^t \Rightarrow e^t q = 5e^t + K \Rightarrow q(t) = 5 + Ke^{-t}$. By continuity at $t = 2$: $q(2) = 5 + Ke^{-2} = 0 \Rightarrow K = -5e^2$. Thus for $t \geq 2$: $q(t) = 5(1 - e^{-(t-2)})$. Expressed with unit step notation.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the structure of the general solution to a linear nonhomogeneous equation $L[y] = g(x)$?

a. $y(x) = y_c(x) + y_p(x)$**, where** y_c **is the complementary solution and** y_p **is any particular solution** ✓

b. $y(x) = y_c(x) \cdot y_p(x)$

c. $y(x) = y_p(x)$ only

d. $y(x) = c_1 y_p(x) + c_2 y_c(x)$

By linearity, the general solution of a nonhomogeneous linear equation is the sum of the complementary solution y_c to the homogeneous equation and any particular solution y_p.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find a particular solution y_p to $y'' - 5y' + 6y = 12$.

a. $y_p = 2$ ✓

b. $y_p = 12$

c. $y_p = 2x$

d. $y_p = 6$

Try constant $y_p = A$. Then $y_p' = 0, y_p'' = 0$. Substituting: $6A = 12 \Rightarrow A = 2$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the trial particular solution y_p for $y'' + 4y = 3x^2 - 1$?

a. $y_p = Ax^2 + Bx + C$ ✓

b. $y_p = Ax^2$

c. $y_p = Ax^2 + B$

d. $y_p = Ax^3 + Bx^2$

For a quadratic polynomial forcing term $g(x) = 3x^2 - 1$, the standard trial solution is a complete quadratic polynomial $y_p = Ax^2 + Bx + C$.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the correct undetermined coefficients form for $y'' + 9y = 5\cos(2x)$?

a. $y_p = A\cos(2x) + B\sin(2x)$ ✓

b. $y_p = A\cos(2x)$

c. $y_p = Ax\cos(2x)$

d. $y_p = A\sin(2x)$

Because $\cos(2x)$ produces both sines and cosines upon differentiation, the complete form is $A\cos(2x) + B\sin(2x)$.

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25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

When must a trial particular solution y_p be multiplied by x (or x^2)?

a. When any term in the trial solution is already a solution of the associated homogeneous equation ✓

b. Whenever the forcing function is a polynomial

c. When the coefficient of y' is negative

d. Only when the right-hand side is zero

If a term in the proposed y_p is part of the complementary solution y_c, substituting it will yield zero; multiplying by x restores linear independence.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For $y'' - 4y' + 4y = 3e^{2x}$, the homogeneous solution is $y_c = (c_1 + c_2x)e^{2x}$. What is the correct trial form for y_p?

a. $y_p = Ax^2e^{2x}$ ✓

b. $y_p = Ae^{2x}$

c. $y_p = Axe^{2x}$

d. $y_p = Ax^3e^{2x}$

Both e^{2x} and xe^{2x} are in y_c. Multiplying by x^2 produces $y_p = Ax^2e^{2x}$, which is linearly independent of y_c.

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What physical phenomenon occurs when an undamped mechanical system is driven at its natural resonance frequency ($y'' + \omega_0^2 y = F_0\cos(\omega_0 t)$)?

a. Resonance, where amplitude grows linearly with time ($y_p = Ct\sin(\omega_0 t)$) ✓

b. Immediate decay to zero

c. Critical damping

d. Chaotic motion

Because the forcing frequency equals the natural frequency, the particular solution takes the form $t\sin(\omega_0 t)$, producing unbounded linearly growing oscillations (resonance).

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28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

If y_{p1} solves $L[y] = g_1(x)$ and y_{p2} solves $L[y] = g_2(x)$, what solves $L[y] = 2g_1(x) - 3g_2(x)$?

a. $2y_{p1} - 3y_{p2}$ ✓

b. $y_{p1}y_{p2}$

c. $\frac{y_{p1}}{y_{p2}}$

d. $y_{p1} + y_{p2}$

By the linearity of differential operators, $L[2y_{p1} - 3y_{p2}] = 2L[y_{p1}] - 3L[y_{p2}] = 2g_1(x) - 3g_2(x)$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Can the Method of Undetermined Coefficients be used to solve $y'' + y = \tan x$?

a. **No, because derivatives of $\tan x$ produce infinitely many non-terminating forms; Variation of Parameters is required** ✓

b. Yes, with trial solution $A\tan x + B\sec x$

c. Yes, with trial solution $A\sin x/\cos x$

d. Only if x is small

Undetermined coefficients applies only to functions with finite dimensional spaces of derivatives (polynomials, exponentials, sines/cosines). For $\tan x$, variation of parameters must be used.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

An undamped oscillator satisfies $y'' + 4y = 3\cos 2t$ (forcing at exactly the natural frequency), $y(0) = 0$, $y'(0) = 0$. Find $y(\pi/4)$.

a. $-3\pi/16$

b. $3\pi/16$ ✓

c. $3\pi/8$

d. $\pi/4$

Homogeneous: $r^2 + 4 = 0 \Rightarrow y_c = C_1\cos 2t + C_2\sin 2t$. The forcing frequency matches the natural frequency exactly (resonance), so the trial needs t: $y_p = t(A\cos 2t + B\sin 2t)$. Substituting and matching gives $A = 0$, $B = \frac{3}{4}$, so $y_p = 34t2t$. With both conditions zero, $y_c \equiv 0$. So $y = 34t2t$, and $y(\pi/4) = \frac{3}{4} \cdot \frac{\pi}{4} \cdot \sin\frac{\pi}{2} = \frac{3\pi}{16}$ — the resonant amplitude $34t$ grows without bound as $t \to \infty$, unlike a non-resonant driven system.

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Find a particular solution of $y'' - y' - 6y = e^{2x} + 18$.

a. $14e^2x - 3$

b. $-14e^2x - 3$ ✓

c. $-14xe^2x - 3$

d. $-14e^2x + 3$

Split the forcing (the superposition note): for e^{2x}, $r = 2$ isn't a root of $r^2 - r - 6 = 0$ (roots $3, -2$), so trial Ae^{2x} gives $(4 - 2 - 6)A = 1 \Rightarrow -4A = 1 \Rightarrow A = -\frac{1}{4}$. For the constant 18, trial B gives $-6B = 18 \Rightarrow B = -3$. Adding the two pieces: $y_p = -14e^2x - 3$.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A plane region R is bounded by the graph of $y = e^{-x^2}$, the x-axis, and the lines $x = 0$ and $x = 1$. A solid is generated by revolving the region R about the line $y = -2$. The cross-sections of the solid perpendicular to the x-axis are squares. The volume of this solid is given by $V = \int_a^b A(x)dx$. (a) Find the outer radius and inner radius of a representative disk if the region were revolved about the line $y = -2$ using the washer method. (b) Set up, but do not evaluate, an integral expression for the volume of the solid generated by revolving the region R about the line $y = -2$ using the washer method. (c) Set up, but do not evaluate, an integral expression for the volume of the solid generated by taking cross-sections perpendicular to the x-axis that are squares. (d) Explain why the integral expression in part (c) is the correct representation for the volume of the solid.

a. The radius of a disk in the washer method is $R(x) = e^{-x^2} + 2$ and $r(x) = 2$.

b. The volume using the washer method is $V = \int_0^1 \pi((e^{-x^2} + 2)^2 - 2^2)dx$.

c. The area of a square cross-section is $A(x) = (e^{-x^2})^2$.

d. The integral expression in part (c) represents the volume because it sums the areas of square cross-sections perpendicular to the x-axis over the interval [0, 1]. ✓

(a) When revolving the region R about the line $y = -2$, the outer radius $R(x)$ is the distance from $y = -2$ to the upper boundary of the region $y = e^{-x^2}$, which is $R(x) = e^{-x^2} - (-2) = e^{-x^2} + 2$. The inner radius $r(x)$ is the distance from $y = -2$ to the lower boundary of the region (the x-axis, $y = 0$), which is $r(x) = 0 - (-2) = 2$. (b) Using the washer method, the volume V is given by the integral of the area of the washers, which is $\pi(R(x)^2 - r(x)^2)$. Therefore, the integral express.

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33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider a region R in the xy-plane bounded by the graphs of $y = \frac{1}{2}x^2$ and $y = 2x$. A solid has its base in region R, and for each value of x in the interval $[0, 4]$, the cross-section perpendicular to the x-axis is an equilateral triangle with side length $s(x) = 2x - \frac{1}{2}x^2$. Write an integral expression that represents the volume of this solid. You do not need to evaluate the integral.

a. $V = \int_0^4 \frac{\sqrt{3}}{4}\left(2x - \frac{1}{2}x^2\right)^2 dx$ ✓

b. $V = \int_0^4 \frac{\sqrt{3}}{4}\left(2x - \frac{1}{2}x^2\right) dx$

c. $V = \int_0^4 \frac{1}{2}\left(2x - \frac{1}{2}x^2\right)^2 dx$

d. $V = \int_0^4 \frac{\sqrt{3}}{2}\left(2x - \frac{1}{2}x^2\right) dx$

The base of the solid is defined by the region R bounded by $y = \frac{1}{2}x^2$ and $y = 2x$. The upper bound is $y = 2x$ and the lower bound is $y = \frac{1}{2}x^2$. The intersection points are found by setting $\frac{1}{2}x^2 = 2x$, which gives $x^2 = 4x$, so $x^2 - 4x = 0$, and $x(x - 4) = 0$. Thus, the region extends from $x = 0$ to $x = 4$. The side length of the equilateral triangle cross-section is given by $s(x) = 2x - \frac{1}{2}x^2$. The area of an equilateral triangle with side length s is.

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The graph of a differentiable function f is shown below. The function f has a local maximum at $x = a$ and a local minimum at $x = b$. The tangent line to the graph of f at $x = c$ has the equation $y - f(c) = \frac{1}{2}(x - c)$. Given that $f(c) = 3$ and $f'(c) = \frac{1}{2}$, find the value of c.

a. 4

b. 3

c. **2** ✓

d. 1

The equation of the tangent line to the graph of a function f at $x = c$ is given by $y - f(c) = f'(c)(x - c)$. We are given that the equation of the tangent line at $x = c$ is $y - f(c) = \frac{1}{2}(x - c)$. Comparing this to the general form, we see that $f'(c) = \frac{1}{2}$. We are also given that $f(c) = 3$. The graph shows a point on the function $f(x)$ at $(x, y) = (2, 3)$. Since the point $(c, f(c))$ must lie on the graph of f, we have $c = 2$ and $f(c) = 3$. Therefore, the value of c is 2.

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35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Consider the function $f(x) = x^2 - 6x + 10$. Find the x-coordinates of the points on the graph of $y = f(x)$ where the tangent line is horizontal.

a. x = 3 ✓

b. x = -3 and x = 3

c. x = 1 and x = 5

d. x = 1

A horizontal tangent line occurs where the derivative of the function is equal to zero. First, find the derivative of $f(x)$: $f'(x) = \frac{d}{dx}(x^2 - 6x + 10) = 2x - 6$. Next, set the derivative equal to zero and solve for x: $2x - 6 = 0$. Adding 6 to both sides gives $2x = 6$. Dividing by 2 gives $x = 3$. Therefore, the x-coordinate of the point where the tangent line is horizontal is $x = 3$.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The graph of the function $f(x)$ is shown below. What is the value of the definite integral $\int_{-2}^{2} f(x)\,dx$?

a. 0

b. 2

c. 4 ✓

d. 8

The definite integral $\int_{-2}^{2} f(x)\,dx$ represents the net signed area between the graph of $f(x)$ and the x-axis from $x = -2$ to $x = 2$. The graph of $f(x)$ for $-2 \le x \le 2$ forms a semicircle above the x-axis with radius 2. The area of this semicircle is $\frac{1}{2}\pi r^2 = \frac{1}{2}\pi(2^2) = 2\pi$. However, the provided figure shows a region that is a combination of triangles and a rectangle. From $x = -2$ to $x = 0$, the area is a triangle with base 2 and height 2, so the area is.

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37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Consider the function $f(x)$ whose derivative is given by $f'(x) = \frac{\sin(x)}{x^2+1}$. A related function $g(x)$ is defined by $g(x) = \int_0^x f(t)dt$. If $f(0) = k$, where k is a positive constant, find the value of $g'(0)$ and determine if $g(x)$ is increasing or decreasing at $x = 0$.

a. $g'(0) = k$**, and** $g(x)$ **is increasing.** ✓

b. $g'(0) = 0$, and $g(x)$ is neither increasing nor decreasing.

c. $g'(0) = 1$, and $g(x)$ is increasing.

d. $g'(0) = k$, and $g(x)$ is decreasing.

We are given $g(x) = \int_0^x f(t)dt$. By the Fundamental Theorem of Calculus Part 1, the derivative of $g(x)$ is $g'(x) = f(x)$. We need to find $g'(0)$. Substituting $x = 0$ into $g'(x)$, we get $g'(0) = f(0)$. We are given that $f(0) = k$, where k is a positive constant. Therefore, $g'(0) = k$. To determine if $g(x)$ is increasing or decreasing at $x = 0$, we examine the sign of $g'(0)$. Since k is a positive constant, $g'(0) = k > 0$. A function is increasing at a point if its derivative at that point is posit.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Separable Equations

An AP Calculus BC FRQ models a population $P(t)$ satisfying the logistic differential equation $\frac{dP}{dt} = \frac{3}{600}P(600 - P)$. At what population size P is the population growing most rapidly?

a. $P = 300$ ✓

b. $P = 600$

c. $P = 150$

d. $P = 0$

In a logistic growth model dP/dt = kP(M - P), the maximum rate of growth occurs at the inflection point where P = M / 2 = 600 / 2 = 300.

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 2.01 Linear First-Order Equations & the Integrating Factor

On an AP Calculus BC Free-Response Question, a particle moves along a curve with velocity vector $\vec{v}(t) = \langle 2t, 3t^2 \rangle$. What is the speed of the particle at time $t = 1$?

a. $\sqrt{13}$ ✓

b. 5

c. $\sqrt{5}$

d. 13

Speed is the magnitude of the velocity vector: ||v(1)|| = sqrt((2*1)^2 + (3*1^2)^2) = sqrt(13).

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 3.01 Second-Order Linear Homogeneous Equations

In a Taylor series BC FRQ, the alternating series $\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k^3}$ is approximated by its first 2 terms S_2. According to the Alternating Series Error Bound, what is the maximum possible error $|S - S_2|$?

a. $\frac{1}{3^3} = \frac{1}{27}$ ✓

b. $\frac{1}{2^3} = \frac{1}{8}$

c. $\frac{1}{3^2} = \frac{1}{9}$

d. $\frac{1}{2 \cdot 3^3}$

By the Alternating Series Error Bound, |S - S_n| <= a_{n+1} = 1 / (2+1)^3 = 1 / 3^3 = 1 / 27.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.01 Variation of Parameters

Consider the differential equation $y'' - xy = 0$. The point $x = 0$ is an ordinary point. If we assume a power series solution of the form $y(x) = \sum_{n=0}^{\infty} c_n x^n$, find the recurrence relation for the coefficients c_n.

a. $c_{n+2} = \frac{c_{n-1}}{(n+2)(n+1)}$ **for** $n \geq 1$ ✓

b. $c_{n+2} = \frac{c_n}{(n+2)(n+1)}$ for $n \geq 0$

c. $c_{n+2} = -\frac{c_{n-1}}{(n+2)(n+1)}$ for $n \geq 1$

d. $c_{n+2} = -\frac{c_n}{(n+2)(n+1)}$ for $n \geq 0$

We are given the differential equation $y'' - xy = 0$ and assume a power series solution of the form $y(x) = \sum_{\infty}^{\infty} c_n x^n$. ; Differentiating twice with respect to x, we get: $y'(x) = \sum_{n=1}^{\infty} n c_n x^{n-1}$ and $y''(x) = \sum_{\infty}^{\infty} n(n-1) c_n x^{n-2}$. ; Substituting these into the differential equation: $\sum_{n=2}^{\infty} n(n-1) c_n x^{n-2} - x \sum_{n=0}^{\infty} c_n x^n = 0$. ; We can rewrite the second term as $\sum_{n=0}^{\infty} c_n x^{n+1}$. To combine the series, we n.

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42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.01 Separable Equations

Consider the system of linear differential equations: $\frac{dx}{dt}, = 3x + y; \frac{dy}{dt}, = x + 3y$ Find the eigenvalues of the coefficient matrix for this system.

a. $\lambda_1 = 2, \lambda_2 = 4$ ✓

b. $\lambda_1 = 1, \lambda_2 = 5$

c. $\lambda_1 = 3, \lambda_2 = 3$

d. $\lambda_1 = -2, \lambda_2 = 4$

The given system of linear differential equations can be written in matrix form as $\mathbf{x}' = A\mathbf{x}$, where $A = (3, 1; 1, 3)$. To find the eigenvalues, we solve the characteristic equation $\det(A - \lambda I) = 0$, where I is the identity matrix and λ represents the eigenvalues. So, $\det((3 - \lambda, 1; 1, 3 - \lambda)) = 0$. This expands to $(3 - \lambda)(3 - \lambda) - (1)(1) = 0$. $(3 - \lambda)^2 - 1 = 0$ $9 - 6\lambda + \lambda^2 - 1 = 0$.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 2.01 Linear First-Order Equations & the Integrating Factor

Consider the differential equation $y''(t) + 4y(t) = u(t - 2)$, where $u(t)$ is the Heaviside step function, with initial conditions $y(0) = 0$ and $y'(0) = 0$. Apply the Laplace Transform to both sides of the equation. If $Y(s) = \mathcal{L}\{y(t)\}$ and $\mathcal{L}\{u(t - a)\} = \frac{e^{-as}}{s}$, which of the following expressions represents $Y(s)$?

a. $\frac{e^{-2s}}{s(s^2 + 4)}$ ✓

b. $\frac{1}{(s^2 + 4)}$

c. $\frac{1}{s(s^2 + 4)}$

d. $\frac{e^{-2s}}{s^2 + 4}$

Applying the Laplace Transform to the given differential equation: $\mathcal{L}\{y''(t)\} + 4\mathcal{L}\{y(t)\} = \mathcal{L}\{u(t - 2)\}$. Using the properties of the Laplace Transform and the given initial conditions $y(0) = 0$ and $y'(0) = 0$, we have $\mathcal{L}\{y''(t)\} = s^2Y(s) - sy(0) - y'(0) = s^2Y(s)$. The Laplace Transform of $y(t)$ is $Y(s)$. The Laplace Transform of the Heaviside step function $u(t - 2)$ is $\frac{e^{-2s}}{s}$. Substituting these into the transformed equation gives.

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Second-Order Linear Homogeneous Equations

Consider the differential equation $\frac{dy}{dx} = x + y$ with the initial condition $y(0) = 1$. Use the Euler method with a step size $h = 0.1$ to approximate the value of $y(0.2)$.

a. 1.20

b. **1.21** ✓

c. 1.23

d. 1.24

The Euler method approximates the solution to an initial value problem $\frac{dy}{dx} = f(x, y)$, $y(x_0) = y_0$ using the iterative formula $y_{n+1} = y_n + hf(x_n, y_n)$, where h is the step size and $x_{n+1} = x_n + h$. Given $f(x, y) = x + y$, $x_0 = 0$, $y_0 = 1$, and $h = 0.1$. For the first step (approximating y(0.1)): $x_1 = x_0 + h = 0 + 0.1 = 0.1$ $y_1 = y_0 + hf(x_0, y_0) = 1 + 0.1(0 + 1) = 1 + 0.1(1) = 1.1$. For the second step (approximating y(0.2)): $x_2 = x_1 + h = 0.1 + 0.1 = 0.2$.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Variation of Parameters

Consider the ordinary differential equation $\frac{dy}{dx} = x + y$, with the initial condition $y(0) = 1$. We will use Euler's method to approximate the value of $y(0.2)$ using a step size of $h = 0.1$. Let $y_0 = y(x_0)$, $y_1 = y(x_1)$, and $y_2 = y(x_2)$ be the approximations at $x_0 = 0$, $x_1 = 0.1$, and $x_2 = 0.2$ respectively. Which of the following statements is TRUE regarding these approximations?

a. $y_1 = 1.1$ and the approximation for $y(0.2)$ is 1.21

b. $y_1 = 1.1$ **and the approximation for** $y(0.2)$ **is** 1.22 ✓

c. $y_1 = 1.2$ and the approximation for $y(0.2)$ is 1.32

d. $y_1 = 1.2$ and the approximation for $y(0.2)$ is 1.30

Euler's method formula is given by $y_{n+1} = y_n + h \cdot f(x_n, y_n)$, where $f(x, y) = x + y$ and $h = 0.1$. For the first step, from $x_0 = 0$ to $x_1 = 0.1$: $y_1 = y_0 + h \cdot f(x_0, y_0)$ $y_1 = 1 + 0.1 \cdot (0 + 1)$ $y_1 = 1 + 0.1 \cdot 1$ $y_1 = 1 + 0.1 = 1.1$ So, the approximation for $y(0.1)$ is $y_1 = 1.1$. For the second step, from $x_1 = 0.1$ to $x_2 = 0.2$: $y_2 = y_1 + h \cdot f(x_1, y_1)$ $y_2 = 1.1 + 0.1 \cdot (0.1 + 1.1)$ $y_2 = 1.1 + 0.1 \cdot (1.2)$ $y_2 = 1.1 + 0.12 = 1.22$ Thus, the appro.

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